



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: POSOCO/NLDC/SO/Daily PSP Report

दिनांक: 23.07.2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई - 400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापलंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 22.07.2024.

महोदय/Dear Sir,

आईईजीसी-2010 की धारा स.-5.5.1 के प्रावधान के अनुसार, दिनांक 22-जुलाई-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 5.5.1 of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 22.07.2024, is available at the NLDC website.

धन्यवाद,
Thanks

Report for previous day

Date of Reporting: 23-Jul-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 20:00 hrs; from RLDCs)	80534	56963	44567	28035	3343	213442
Peak Shortage (MW)	175	719	0	100	0	994
Energy Met (MU)	1901	1340	1014	620	62	4937
Hydro Gen (MU)	396	35	113	71	38	653
Wind Gen (MU)	16	149	280	-	-	445
Solar Gen (MU)*	148.07	58.14	90.81	1.51	0.66	299
Energy Shortage (MU)	1.43	2.05	0.00	0.69	0.00	4.17
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	83845	58530	48008	29211	3351	217922
Time Of Maximum Demand Met	12:11	11:22	10:27	23:19	19:40	11:52

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.048	0.31	1.44	10.06	11.80	72.31	15.89

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	15420	0	365.5	221.3	-1.7	142	0.00
	Haryana	13081	0	281.1	207.7	-1.8	347	0.00
	Rajasthan	16319	0	340.1	128.0	1.2	839	0.46
	Delhi	6922	0	148.2	124.8	-1.0	260	0.00
	UP	29682	0	611.5	286.9	-0.3	577	0.00
	Uttarakhand	2246	0	50.5	28.1	0.7	337	0.37
	HP	1788	0	37.9	0.5	-0.1	99	0.00
	J&K(UT) & Ladakh(UT)	2651	250	54.4	27.9	0.6	411	0.60
	Chandigarh	402	0	8.1	8.0	0.1	45	0.00
WR	Railways_NR ISTS	210	0	4.2	3.7	0.5	80	0.00
	Chhattisgarh	5008	0	114.7	69.0	-1.3	240	0.00
	Gujarat	19080	0	412.1	144.4	0.0	750	0.00
	MP	11805	0	259.2	148.9	-1.5	855	2.05
	Maharashtra	22325	0	481.8	152.3	-6.5	804	0.00
	Goa	622	0	12.6	12.8	-0.3	94	0.00
	DNHDDPDCL	1306	0	29.5	29.5	0.0	50	0.00
	AMNSIL	854	0	17.3	5.2	0.1	269	0.00
	BALCO	529	0	12.6	12.6	0.0	10	0.00
SR	Andhra Pradesh	8781	0	188.3	22.7	-1.1	472	0.00
	Telangana	9762	0	193.6	102.6	2.5	749	0.00
	Karnataka	9912	0	189.6	-32.0	-1.7	994	0.00
	Kerala	3762	0	76.3	43.6	0.1	454	0.00
	Tamil Nadu	16804	0	356.5	157.8	0.6	1228	0.00
	Puducherry	475	0	10.1	9.6	0.1	77	0.00
ER	Bihar	7781	0	161.0	151.7	-1.1	452	0.00
	DVC	3426	0	74.0	-37.8	-0.6	301	0.00
	Jharkhand	2302	0	45.4	32.5	1.5	286	0.69
	Odisha	5263	0	112.9	38.1	-3.3	403	0.00
	West Bengal	10510	0	225.0	88.0	-1.8	257	0.00
	Sikkim	83	0	1.3	1.3	0.0	37	0.00
	Railways_ER ISTS	15	0	0.2	0.2	0.0	0	0.00
NER	Arunachal Pradesh	157	0	2.9	2.7	-0.3	25	0.00
	Assam	2242	0	41.8	34.5	0.1	137	0.00
	Manipur	180	0	1.4	3.0	-1.6	2	0.00
	Meghalaya	306	0	5.4	-0.6	-0.2	59	0.00
	Mizoram	118	0	1.9	0.9	-0.4	0	0.00
	Nagaland	188	0	3.0	2.6	-0.1	20	0.00
	Tripura	306	0	5.5	5.9	-0.3	43	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	40.3	16.3	-24.1	-22.1
Day Peak (MW)	1941.0	791.9	-1023.0	-1115.0

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	443.1	-267.0	-93.2	-79.2	-3.6	0.0
Actual(MU)	446.6	-267.8	-100.9	-78.4	-7.1	-7.4
O/D/U/D(MU)	3.6	-0.7	-7.6	0.9	-3.5	-7.4

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	3207	12461	7238	4716	280	27902	53
State Sector	3911	10130	9283	1574	157	25054	47
Total	7118	22591	16521	6290	437	52956	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	899	1377	544	698	15	3533	67
Lignite	30	10	49	0	0	90	2
Hydro	396	35	113	71	38	653	12
Nuclear	30	63	76	0	0	168	3
Gas, Naptha & Diesel	34	32	5	0	23	95	2
RES (Wind, Solar, Biomass & Others)	171	210	392	4	1	778	15
Total	1560	1728	1179	772	77	5316	100

Share of RES in total generation (%)	10.98	12.16	33.26	0.47	0.87	14.65
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	38.29	17.82	49.27	9.69	50.20	30.03

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.023
Based on State Max Demands	1.067

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	217922	11:52	0
Non-Solar hr	215164	19:50	767

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 07:00 to 17:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 23-Jul-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	702	0.0	15.8	-15.8
2	HVDC	PUSAULI B/B	-	0	95	0.0	2.4	-2.4
3	765 kV	GAYA-VARANASI	2	627	716	0.0	6.4	-6.4
4	765 kV	SASARAM-FATEHPUR	1	271	298	0.0	2.9	-2.9
5	765 kV	GAYA-BALIA	1	0	790	0.0	14.2	-14.2
6	400 kV	PUSAULI-VARANASI	1	0	113	0.0	1.3	-1.3
7	400 kV	PUSAULI -ALLAHABAD	1	20	62	0.0	1.1	-1.1
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	0	815	0.0	12.9	-12.9
9	400 kV	PATNA-BALIA	2	112	726	0.0	11.5	-11.5
10	400 kV	NAUBATPUR-BALIA	2	99	203	0.0	3.0	-3.0
11	400 kV	BIHARSHARIFF-BALIA	2	240	191	0.0	1.0	-1.0
12	400 kV	MOTIHARI-GORAKHPUR	2	3	440	0.0	7.2	-7.2
13	400 kV	BIHARSHARIFF-VARANASI	2	0	0	0.0	0.0	0.0
14	220 kV	SAHUPURI-KARAMNASA	1	0	75	0.0	1.6	-1.6
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.0	0.0	0.0
16	132 kV	GARWAH-RIHAND	1	30	0	0.5	0.0	0.5
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						0.5	81.4	-80.9
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	0	1458	0.0	19.1	-19.1
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	1907	0	35.1	0.0	35.1
3	765 kV	JHARSUGUDA-DURG	2	208	605	0.0	4.2	-4.2
4	400 kV	JHARSUGUDA-RAIGARH	4	0	531	0.0	7.8	-7.8
5	400 kV	RANCHI-SIPAT	2	494	124	5.0	0.0	5.0
6	220 kV	BUDHIPADAR-RAIGARH	1	25	104	0.0	1.0	-1.0
7	220 kV	BUDHIPADAR-KORBA	2	0	107	0.0	1.3	-1.3
ER-WR						40.1	33.3	6.8
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	756	0	8.6	0.0	8.6
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	794	0.0	13.0	-13.0
3	765 kV	ANGUL-SRIKAKULAM	2	0	2190	0.0	25.6	-25.6
4	400 kV	TALCHER-I/C	2	1115	42	18.4	0.0	18.4
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						8.6	38.5	-29.9
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	181	155	0.5	1.2	-0.7
2	400 kV	ALIPURDUAR-BONGAIGAON	2	165	315	0.0	1.2	-1.2
3	220 kV	ALIPURDUAR-SALAKATI	2	18	83	0.0	0.6	-0.6
ER-NER						0.5	2.9	-2.5
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	0	508	0.0	12.1	-12.1
NER-NR						0.0	12.1	-12.1
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	5031	0.0	111.4	-111.4
2	HVDC	VINDHYACHAL B/B	-	0	309	0.0	2.0	-2.0
3	HVDC	MUNDRA-MOHINDERGARH	2	0	2358	0.0	60.4	-60.4
4	765 kV	GWALIOR-AGRA	2	0	2631	0.0	40.8	-40.8
5	765 kV	GWALIOR-PHAGI	2	490	1570	1.3	22.6	-21.4
6	765 kV	JABALPUR-ORAI	2	0	1431	0.0	48.2	-48.2
7	765 kV	GWALIOR-ORAI	1	753	0	13.3	0.0	13.3
8	765 kV	SATNA-ORAI	1	0	1000	0.0	20.0	-20.0
9	765 kV	BANASKANTHA-CHITORGARH	2	985	1268	4.6	10.7	-6.1
10	765 kV	VINDHYACHAL-VARANASI	2	0	3894	0.0	65.2	-65.2
11	400 kV	ZERDA-KANKROLI	1	219	210	1.2	1.7	-0.6
12	400 kV	ZERDA -BHINMAL	1	151	145	0.8	1.2	-0.4
13	400 kV	VINDHYACHAL -RIHAND	1	972	0	19.8	0.0	19.8
14	400 kV	RAPP-SHUJALPUR	2	267	560	1.0	6.7	-5.7
15	400 kV	NEEMUCH-Chittorgarh	2	310	591	1.2	6.8	-5.6
16	220 kV	BHANPURA-RANPUR	1	0	109	0.0	2.0	-2.0
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	2.1	-2.1
18	220 kV	MEHGAON-AURAIYA	1	94	2	0.9	0.0	0.9
19	220 kV	MALANPUR-AURAIYA	1	63	28	0.4	0.1	0.3
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						44.4	401.9	-357.5
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	996	0	23.9	0.0	23.9
2	HVDC	RAIGARH-PUGALUR	2	771	0	12.5	0.0	12.5
3	765 kV	SOLAPUR-RAICHUR	2	2570	0	33.4	0.0	33.4
4	765 kV	WARDHA-NIZAMABAD	2	535	1718	2.2	9.4	-7.2
5	765 kV	WARORA-WARANGAL(NEW)	2	541	1810	2.1	9.9	-7.9
6	400 kV	KOLHAPUR-KUDGI	2	2288	0	40.8	0.0	40.8
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	66	0.6	0.0	0.6
9	220 kV	XELDEM-AMBEWADI	1	1	109	1.1	0.0	1.1
WR-SR						116.6	19.4	97.3

INTERNATIONAL EXCHANGES					Import(+ve)/Export(-ve)		
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)	
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	728	0	656	15.75	
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	932	854	858	20.58	
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	133	65	84	2.02	
	NER	132kV GELEPHU-SALAKATI	34	22	30	0.72	
	NER	132kV MOTANGA-RANGIA	66	31	50	1.21	
NEPAL	NR	NEPAL IMPORT (FROM UP)	0	0	0	0.00	
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	66	0	51	1.22	
	ER	NEPAL IMPORT (FROM BIHAR)	0	0	0	0.00	
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	726	269	628	15.08	
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-937	-924	-930	-22.31	
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-1115	-809	-923	-22.14	
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-86	0	-75	-1.80	

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nepal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bangladesh	22.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.21
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	22.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.21

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL	
		BILATERAL TOTAL	COLLECTIVE							
			IDAM			RTM				
			IEX	PXIL	HPX	IEX	PXIL	HPX		
Bhutan	38.36	0.00	1.12	0.00	0.00	0.00	0.00	0.00	39.48	
Nepal	2.63	0.00	13.31	0.00	0.00	0.00	0.19	0.00	0.00	16.13
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	40.99	0.00	14.43	0.00	0.00	0.00	0.19	0.00	0.00	55.61

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL	
		BILATERAL TOTAL	COLLECTIVE							
			IDAM			RTM				
			IEX	PXIL	HPX	IEX	PXIL	HPX		
Bhutan	38.36	0.00	1.12	0.00	0.00	0.00	0.00	0.00	39.48	
Nepal	2.63	0.00	13.31	0.00	0.00	0.00	0.19	0.00	0.00	16.13
Bangladesh	-22.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-22.21
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	18.78	0.00	14.43	0.00	0.00	0.00	0.19	0.00	0.00	33.40